

- This exam contains 8 pages (including the front and back of this cover page) and 9 problems. Check to see if any pages are missing.
- This is the final exam and will count for 40% of your final grade. You have 120 minutes (1:15 PM - 3:15 PM) to complete this exam. You may use one index card of notes and a calculator on this exam, but no other outside sources may be consulted.

The following rules apply:

- Every problem on this exam, other than true-or-false problems, requires work of some form. An incorrect answer supported by substantially correct calculations and explanations will receive partial credit. **A correct answer that has no supporting work will not receive full credit.**
- Organize your work, in a reasonably neat and coherent way, in the space provided. Work scattered all over the page without a clear ordering will receive very little to no credit.
- There are 440 points total in this exam, as well as 2 bonus questions worth a total of 20 possible points.

|         |    |     |     |    |    |       |
|---------|----|-----|-----|----|----|-------|
| Page:   | 3  | 4   | 5   | 6  | 7  | Total |
| Points: | 40 | 140 | 120 | 60 | 80 | 440   |
| Score:  |    |     |     |    |    |       |

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| Bonus: |  |
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| Review Day Question Bonus: |  |
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1. (a)  If  $\lambda$  is a scalar such that  $Ax = \lambda x$  for some  $x \in \mathbb{R}^n$ ,  $\lambda$  is an eigenvalue of  $A$ .
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- (b)   $\det(AB) = \det(A) - \det(B)$ .
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- (c)  If  $A = P \begin{bmatrix} 3 & 0 \\ 0 & 8 \end{bmatrix} P^{-1}$ , then  $A^{2019} = P \begin{bmatrix} 3^{2019} & 0 \\ 0 & 8^{2019} \end{bmatrix} P^{-1}$ .
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- (d)  If  $A$  is diagonalizable, then  $A$  is invertible.
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- (e)  If a vector  $x$  is orthogonal to every vector in a subspace  $W$ , then  $x \in W^\perp$ .
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- (f)  If  $A$  is any  $n \times m$  matrix, then  $(\text{Col}(A))^\perp = \text{Nul}(A)$ .
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- (g)  If  $\{v_1, \dots, v_p\}$  is a linearly independent set in  $\mathbb{R}^n$ , then it is also an orthogonal set.
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- (h)  If  $\{v_1, \dots, v_p\}$  is an orthogonal basis for a subspace  $W$  of  $\mathbb{R}^n$ , then for any  $y \in W$ , we have  $y = \frac{y \cdot v_1}{v_1 \cdot v_1} v_1 + \dots + \frac{y \cdot v_p}{v_p \cdot v_p} v_p$ .
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- (i)  The best approximation to  $y$  by elements of a subspace  $W$  is given by the vector  $y - \text{proj}_W(y)$ .
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- (j)  If  $x \in \mathbb{R}^n$  is a nonzero vector and  $W$  is a nonzero subspace of  $\mathbb{R}^n$ , then  $\text{proj}_W(x) \neq 0$ .
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2. 60 points Show that if  $A^2$  is the zero matrix, then the only eigenvalue of  $A$  is 0.

3. 80 points Find the characteristic polynomial and the eigenvalues of the matrix  $\begin{bmatrix} -3 & 4 \\ 6 & 2 \end{bmatrix}$ .

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4. 60 points Suppose  $A$  has  $n$  linearly independent eigenvectors. Prove that  $A^T$  also has  $n$  linearly independent eigenvectors. (*HINT: Use the Diagonalization Theorem*)
5. 60 points Let  $T : \mathbb{P}_2 \rightarrow \mathbb{P}_1$  be the transformation that maps a polynomial,  $p(t)$ , to its second derivative,  $p''(t)$ . Find the matrix for  $T$  relative to the basis  $\mathcal{B} = \{1, t, t^2\}$  for  $\mathbb{P}_2$  and the basis  $\mathcal{D} = \{1, t\}$  for  $\mathbb{P}_1$ .

6. 60 points Let  $v_1 = \begin{bmatrix} 2 \\ -1 \\ -3 \\ 1 \end{bmatrix}$ ,  $v_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ -1 \end{bmatrix}$ , and  $z = \begin{bmatrix} 3 \\ -7 \\ 2 \\ 3 \end{bmatrix}$ . Let  $W = \text{span}\{v_1, v_2\}$  and find the best approximation to  $z$  by elements of  $W$ .

7. 80 points Let  $v_1 = \begin{bmatrix} 3 \\ 0 \\ -1 \end{bmatrix}$  and  $v_2 = \begin{bmatrix} 8 \\ 5 \\ 6 \end{bmatrix}$ , and write  $W = \text{span}\{v_1, v_2\}$ . Use the Gram-Schmidt process to find an orthogonal basis for  $W$ .

8. 10 points (bonus) Let  $W$  be a subspace of  $\mathbb{R}^n$ . Show that if  $x \in \mathbb{R}^n$  is a vector which satisfies both  $x \in W$  and  $x \in W^\perp$ , then  $x = 0$ .

9. 10 points (bonus) Let  $A$  be an  $m \times n$  matrix such that  $A^T A$  is invertible. Show that the columns of  $A$  are linearly independent.

(WARNING: Since  $A$  may not be a square matrix, you may **not** assume that  $A$  is invertible!)